

From atomic entities to physico-chemical systems

Worksheets

5 units · 15 questions each, from recall to challenge

Semester 1 · Academic Year 2026–2027

How to use the worksheets

- Revise the unit (study guide: concepts and recap) before starting its worksheet.
- Work on paper, without looking at the answers. Write the formula with letters first, then the numbers with their units.
- The questions get harder as you go. Aim for all of Levels 1–2 before attempting Level 3; the Challenge questions link several ideas or go slightly beyond the handout.
- Check yourself with *Chem-S1-1__Worksheet-Answers.pdf*.

Levels

Level 1 •

Q1–4: recall a definition, apply one formula.

Level 2 ••

Q5–8: explain a result or chain two steps.

Level 3 •••

Q9–12: multi-step problems, interpreting data.

Challenge ★

Q13–15: synthesis, links between units, going beyond the handout.

Contents

Worksheet	Unit
1	Classical description of the atom
2	Quantum model of the atom
3	Quantum numbers and atomic orbitals
4	Electronic configuration and periodic table
5	Atomic properties and periodicity

Worksheet 1 · Classical description of the atom

Data $e = 1.6 \times 10^{-19} \text{ C}$ $m_e = 9.1094 \times 10^{-31} \text{ kg}$ $m_p = 1.6727 \times 10^{-27} \text{ kg}$ $m_n = 1.6749 \times 10^{-27} \text{ kg}$
 $1 \text{ amu} = 1.6605 \times 10^{-27} \text{ kg}$ $N_A = 6.022 \times 10^{23} / \text{mol}$ $c = 3.00 \times 10^8 \text{ m/s}$ $k = 1/(4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. Calculator allowed.

Q1. Level 1 • Who discovered what?

Match each scientist to the date and to the discovery or idea: Democritus, Dalton, Thomson, Millikan, Rutherford. Discoveries: (a) the nucleus; (b) the elementary charge; (c) indivisible atoms; (d) the electron; (e) compounds contain fixed ratios of atoms of given masses.

Q2. Level 1 • Counting particles

Give the number of protons, neutrons and electrons of $^{12}_6\text{C}$, $^{14}_6\text{C}$, $^{35}_{17}\text{Cl}$ and $^{56}_{26}\text{Fe}$. Which two are isotopes?

Q3. Level 1 • Units of length

Express a typical atomic size, 100 pm, in m, nm and Å. Express 1 fm in pm. What is the typical ratio between the size of an atom and the size of its nucleus?

Q4. Level 1 • Cathode rays

In Thomson's tube the beam bends towards the positively charged plate. What does this show? Why was it important that the same deflection was obtained with any gas and any metal?

Q5. Level 2 •• Ions and isoelectronic species

Give the number of protons, neutrons and electrons of $^{24}_{12}\text{Mg}^{2+}$, $^{27}_{13}\text{Al}^{3+}$, $^{19}_9\text{F}^-$ and $^{40}_{20}\text{Ca}^{2+}$. Which species are isoelectronic, and with which noble gas?

Q6. Level 2 •• Reading Rutherford's experiment

Give the three observations of the gold-foil experiment and, for each one, what it tells us about the atom. Why could Thomson's "plum pudding" model not explain them?

Q7. Level 2 •• How heavy are electrons?

(a) Compute m_p/m_e . (b) What fraction of the mass of a ^{12}C atom (exactly 12 amu) is due to its electrons? Conclude.

Q8. Level 2 •• A stadium-sized atom

Take an atom of diameter 100 pm and a nucleus of diameter 10^{-3} pm. If the nucleus were enlarged to a marble of diameter 1.0 cm, what would be the diameter of the atom? What fraction of the atom's volume does the nucleus occupy?

Q9. Level 3 ••• Charge-to-mass ratios

(a) Compute e/m for the electron and for the proton. (b) An electron and a proton enter the same electric field with the same speed. Which one is deflected more, and by what factor (the deflection is proportional to q/m)? (c) Why did this make the electron easy to discover with cathode rays?

Q10. Level 3 ••• From amu to grams per mole

One atom of gold has a mass of 196.967 amu. (a) Express it in kg. (b) How many atoms are there in 1.00 g of gold? (c) How many moles is that? Deduce the molar mass of gold and state the general rule linking amu and g mol^{-1} .

Q11. Level 3 ●●● Identify the species

Identify each species (symbol, A , charge). Use a periodic table for the symbols. (a) 26 protons, 30 neutrons, 23 electrons. (b) 17 protons, 20 neutrons, 18 electrons. (c) A neutral atom with $A = 39$ and one more neutron than protons. (d) An ion with 10 electrons, 8 neutrons and charge $2-$.

Q12. Level 3 ●●● Density of nuclear matter

A ^{12}C nucleus has a mass of about 12 amu and a radius of about 2.7 fm. (a) Compute its density. (b) What would be the mass of a teaspoon (5 cm^3) of nuclear matter? Compare with water (1000 kg/m^3).

Q13. Challenge ★ Millikan's oil drops

Millikan measured the charges of oil drops: 3.2×10^{-19} , 4.8×10^{-19} , 8.0×10^{-19} , 6.4×10^{-19} and $1.12 \times 10^{-18}\text{ C}$. (a) Show that these values are consistent with a single elementary charge and find it. (b) How many extra electrons does each drop carry? (c) Could a drop carry $2.4 \times 10^{-19}\text{ C}$?

Q14. Challenge ★ Where does the missing mass go?

(a) Using the masses in amu of the proton (1.0073), the neutron (1.0087) and the electron ($1/1823$), compute the total mass of the particles that make up a ^{12}C atom. (b) Compare with the mass of the atom, 12 amu exactly: compute the mass defect Δm in amu and in kg. (c) The energy that holds the nucleus together is $E = \Delta m c^2$. Compute it per atom and per mole, and compare with a typical chemical bond energy (about 400 kJ mol^{-1}). Why can chemical reactions not change one element into another?

Q15. Challenge ★ How small is the gold nucleus?

In Rutherford's experiment, an α -particle (charge $+2e$) with kinetic energy 5.0 MeV ($1\text{ MeV} = 1.6 \times 10^{-13}\text{ J}$) heads straight for a gold nucleus ($Z = 79$). It stops and turns back at the distance d where all its kinetic energy has become electric potential energy $E_p = k q_1 q_2 / d$. (a) Compute d . (b) What does this tell you about the size of the nucleus? Compare with the size of the gold atom (about 140 pm).

Worksheet 2 · Quantum model of the atom

Data $c = 3.00 \times 10^8 \text{ m/s}$ $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $N_A = 6.022 \times 10^{23} / \text{mol}$
 $R_H = 3.29 \times 10^{15} \text{ Hz}$ hydrogen: $E_n = -13.6/n^2 \text{ eV}$; hydrogen-like ion: $E_n = -13.6 Z^2/n^2 \text{ eV}$. Calculator allowed.

Q1. Level 1 • Ordering the spectrum

Put these radiations in order of increasing frequency: visible light, X-rays, radio waves, infrared, γ rays, ultraviolet, microwaves. Which one has the longest wavelength? Which photons carry the most energy?

Q2. Level 1 • A red laser

A laser emits red light at $\lambda = 650 \text{ nm}$. Compute its frequency ν and its period T .

Q3. Level 1 • Energy of one photon

Compute the energy of one photon of the laser of Q2, in joules and in electronvolts.

Q4. Level 1 • Vocabulary of the Bohr model

Define ground state, excited state and ionised state for the hydrogen atom, and give the energy of the electron in each case (for the excited state, give E_2 and E_3).

Q5. Level 2 •• Emission versus absorption

Describe the emission spectrum and the absorption spectrum of hydrogen gas in the visible. Explain why the lines appear at the same wavelengths in both.

Q6. Level 2 •• Why is permanganate purple?

An aqueous solution of KMnO_4 absorbs light between about 515 and 565 nm. (a) Which colour is absorbed? What colour does the solution appear? (b) Why is the absorption a broad band and not thin lines as for hydrogen gas?

Q7. Level 2 •• The Balmer lines from Rydberg's formula

Using $\nu = R_H(1/n_1^2 - 1/n_2^2)$ with $n_1 = 2$, compute the wavelengths for $n_2 = 3, 4, 5, 6$ and compare them with the handout values.

Q8. Level 2 •• Rydberg meets Bohr

(a) Using Bohr's levels, compute the energy (in eV and J) of the photon emitted in the transition $3 \rightarrow 2$, and its wavelength. (b) Show that Bohr's formula gives Rydberg's formula and that $R_H = 13.6 \text{ eV}/h$. Check the numerical value.

Q9. Level 3 ••• Series and their limits

(a) Compute the wavelength of the first Lyman line ($2 \rightarrow 1$) and the Lyman series limit. In which region are they? (b) Compute the Paschen series limit ($n_1 = 3$). (c) Explain why only the Balmer series has lines in the visible.

Q10. Level 3 ••• Ionisation energy of hydrogen

(a) What energy is needed to ionise a hydrogen atom in its ground state? Give it in eV, in J per atom and in kJ mol^{-1} . (b) What is the longest wavelength that can do it? (c) Unit 5 lists ionisation energies in kJ mol^{-1} but leaves hydrogen blank: fill it in.

Q11. Level 3 ... Counting emission lines

Hydrogen atoms are excited to $n = 4$ and then return to the ground state by all possible paths. (a) How many different lines can be emitted? List the transitions. (b) Sort them into series and spectral regions. (c) Which ones can you see?

Q12. Level 3 ... Which photons are absorbed?

Hydrogen atoms in their ground state are hit by photons of energy (a) 10.2 eV, (b) 11.0 eV, (c) 12.09 eV, (d) 15.0 eV. For each, say whether it is absorbed and what happens to the atom.

Q13. Challenge ★ Identify the transition

A line of the hydrogen emission spectrum is measured at $\lambda = 1876$ nm. Find n_1 and n_2 . Explain your method.

Q14. Challenge ★ The helium ion

He^+ ($Z = 2$) has a single electron. (a) Give its energy levels E_1 , E_2 , E_3 and its ionisation energy. (b) Compute the wavelength of its $3 \rightarrow 2$ transition; in which region is it? (c) Show that the He^+ transition $6 \rightarrow 4$ has exactly the same wavelength as the red line of hydrogen. Why is this so?

Q15. Challenge ★ Photons in a laser beam

A green laser pointer emits 1.0 mW at 532 nm. (a) How many photons does it emit per second? (b) How many moles of photons per hour? (c) Why do we not notice that light comes in packets?

Worksheet 3 · Quantum numbers and atomic orbitals

Data $c = 3.00 \times 10^8 \text{ m/s}$ $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $a_0 = 52.9 \text{ pm}$ hydrogen-like species: $E_n = -13.6 Z^2/n^2 \text{ eV}$, $r = n^2 a_0/Z$. Calculator allowed.

Q1. Level 1 • The third shell

For $n = 3$, list the allowed values of ℓ with the name of each subshell, and the values of m_ℓ for each. How many orbitals does the shell contain?

Q2. Level 1 • Allowed or not?

Which of these sets (n, ℓ, m_ℓ) are allowed? Name the orbital when it is, and give the reason when it is not: (a) (2, 2, 0); (b) (3, 1, -1); (c) (1, 0, 1); (d) (4, 3, -2); (e) (2, 1, 2); (f) (5, 0, 0).

Q3. Level 1 • Naming wavefunctions

Name the orbital described by $\Psi_{2,0,0}$, $\Psi_{3,1,1}$, $\Psi_{4,2,-2}$ and $\Psi_{4,3,0}$. Conversely, give all the wavefunctions of the 2p subshell.

Q4. Level 1 • Spin and boundary surfaces

(a) Which values can m_s take, and how are they drawn? Which experiment revealed spin? (b) What is a boundary surface? What is the shape of the boundary surface of an s orbital, and of a p orbital?

Q5. Level 2 •• Energies and degeneracy in hydrogen

(a) Give the energy of the electron of hydrogen described by $\Psi_{3,2,-1}$ and by $\Psi_{3,0,0}$. What word describes this situation? (b) How many degenerate orbitals are there in the shell $n = 4$ of hydrogen?

Q6. Level 2 •• Ground or excited?

For the hydrogen atom, say which orbital is occupied, whether the atom is in its ground state or an excited state, and its energy, when the electron is described by (a) $\Psi_{1,0,0}$, (b) $\Psi_{2,1,0}$, (c) $\Psi_{3,2,1}$. (d) How much energy is needed to go from (a) to (b)?

Q7. Level 2 •• Counting nodes

Give the number of radial nodes, angular nodes and total nodes of the 2s, 3p, 3d, 4s and 4f orbitals. Check the 3s, 3p and 3d cases on Figure 3-3 of the handout.

Q8. Level 2 •• The three p orbitals

For each of p_x , p_y and p_z , give the axis along which it points and its nodal plane. Which one corresponds to $m_\ell = 0$? What happens to the sign of the wavefunction from one lobe to the other?

Q9. Level 3 ••• Size of orbitals

Using $r = n^2 a_0/Z$, compute the radius of: (a) the 1s, 2s and 3s orbitals of hydrogen; (b) the 1s orbital of He^+ and of Li^{2+} . (c) Explain in one sentence the effect of n and of Z .

Q10. Level 3 ••• Ordering subshells

Using Madelung's rule, write $n + \ell$ for each subshell and put them in order of increasing energy: 5s, 4f, 3d, 4p, 6s, 4d, 5p, 4s.

Q11. Level 3 ••• Hydrogen versus potassium

In hydrogen the 3d orbitals have the same energy as 3s, and lie far below 4s. In potassium ($Z = 19$) the 4s orbital is lower than 3d. (a) Which approximation lets us still use the labels 1s, 2s, 2p ... for many-electron atoms? (b) Explain why the energy now depends on ℓ and why 4s can drop below 3d.

Q12. Level 3 ●●● **The Zeeman effect**

- (a) Into how many sublevels does a p subshell split in a magnetic field? a d subshell? an f subshell?
(b) Hydrogen atoms emit the $2p \rightarrow 1s$ line (122 nm). How many lines are seen in a magnetic field?
(c) What does the Zeeman effect prove?

Q13. Challenge ★ **How many electrons can a shell hold?**

An electron is described by four quantum numbers (n, ℓ, m_ℓ, m_s) . (a) List all the allowed sets for $n = 2$. (b) Show that the number of sets for a shell n is $2n^2$ and check it for $n = 1, 2, 3$. (Unit 4: no two electrons of an atom can share the same set, so this is the maximum number of electrons of the shell.)

Q14. Challenge ★ **The sodium doublet**

The D line of sodium is a doublet: $\lambda_1 = 589.0$ nm and $\lambda_2 = 589.6$ nm. (a) Compute the energy of each photon in J and in eV (4 significant figures). (b) Compute the energy difference between the two upper levels in J and in meV. (c) Explain the origin of the doublet.

Q15. Challenge ★ **Where is the 1s electron?**

The radial probability density of the 1s orbital of hydrogen is $P(r) = \frac{4}{a_0^3} r^2 e^{-2r/a_0}$. (a) Show that P is maximum at $r = a_0$. (b) Explain why the probability density is zero at $r = 0$ even though $|\Psi_{1s}|^2$ is largest at the nucleus. (c) For He^+ , $P(r) \propto r^2 e^{-2Zr/a_0}$: where is its maximum?

Worksheet 4 · Electronic configuration and periodic table

Data Atomic numbers: H 1, He 2, Li 3, Be 4, B 5, C 6, N 7, O 8, F 9, Ne 10, Na 11, Mg 12, Al 13, Si 14, P 15, S 16, Cl 17, Ar 18, K 19, Ca 20, Sc 21, Ti 22, V 23, Cr 24, Mn 25, Fe 26, Co 27, Ni 28, Cu 29, Zn 30, Ga 31, Ge 32, As 33, Se 34, Br 35, Kr 36, Rb 37, Sr 38. No calculator needed.

Q1. Level 1 • Capacity of subshells

Complete: an s subshell has ... orbital(s) and holds at most ... electrons; p: ... and ...; d: ... and ...; f: ... and Which principle limits an orbital to two electrons?

Q2. Level 1 • The three rules

State the Aufbau principle, the Pauli exclusion principle and Hund's rule in one sentence each.

Q3. Level 1 • Writing configurations

Write the full ground-state configuration of O, Na, Cl and Ar, then the shorthand form with a noble gas core.

Q4. Level 1 • Core and valence

Give the number of core electrons and of valence electrons of C, Mg, S and Ar.

Q5. Level 2 •• Box diagrams and Hund's rule

(a) Draw the box diagram of the valence shell of N, O and F and give the number of unpaired electrons. (b) Which of these $2p^3$ diagrams respects Hund's rule? Explain what is wrong with the others: (i) $\uparrow\downarrow\uparrow\uparrow$ (ii) $\uparrow\uparrow\uparrow\uparrow$ (iii) $\uparrow\uparrow\downarrow\uparrow$.

Q6. Level 2 •• From configuration to element

Identify the element and give its period, group and block: (a) $1s^2 2s^2 2p^6 3s^2 3p^4$; (b) $[\text{Ne}] 3s^1$; (c) $[\text{Ar}] 3d^6 4s^2$; (d) $[\text{Kr}] 5s^2$.

Q7. Level 2 •• From position to configuration

Write the shorthand configuration of the element located at: (a) period 3, group 15; (b) period 4, group 2; (c) period 4, group 17; (d) period 5, group 1. Name each element.

Q8. Level 2 •• Ground, excited or impossible?

Classify each configuration as ground state, excited state or impossible, and name the atom: (a) $1s^2 2s^2 2p^6 3s^1$; (b) $1s^2 2s^2 2p^5 3s^1$; (c) $1s^2 2s^3$; (d) $1s^2 2p^1$; (e) $1s^2 2s^2 2p^7$; (f) $1s^2 2s^2 2p^6 3d^1$.

Q9. Level 3 ••• Ions and noble gases

Write the configuration of Na^+ , Mg^{2+} , O^{2-} , Cl^- and S^{2-} . Which noble gas does each resemble? Explain why these ions are the ones usually formed.

Q10. Level 3 ••• Iron and its ions

(a) Write the configuration of Fe ($Z = 26$) and draw the box diagram of its 3d and 4s subshells. How many unpaired electrons does it have? (b) Write the configuration of Fe^{2+} and Fe^{3+} . (c) Fe^{3+} is particularly stable. Suggest why.

Q11. Level 3 ••• Exceptions: chromium and copper

(a) Write the configuration predicted by Madelung's rule for Cr ($Z = 24$) and Cu ($Z = 29$). (b) The real configurations are $[\text{Ar}] 3d^5 4s^1$ and $[\text{Ar}] 3d^{10} 4s^1$. Explain. (c) How many unpaired electrons do Cr and Cu have?

Q12. Level 3 ●●● **Four quantum numbers**

(a) Give a possible set (n, ℓ, m_ℓ, m_s) for each of the five valence electrons of nitrogen in its ground state. (b) Which of these sets could describe an electron of a ground-state phosphorus atom? (i) $(3, 1, 0, +\frac{1}{2})$; (ii) $(3, 2, 0, +\frac{1}{2})$; (iii) $(2, 1, -1, -\frac{1}{2})$; (iv) $(3, 0, 0, 0)$.

Q13. Challenge ★ **Why 2, 8, 8, 18, 18, 32?**

Using Madelung's order and the fact that each period starts when a new ns subshell starts filling, list the subshells filled in each of the first six periods and deduce the number of elements in each.

Q14. Challenge ★ **Magnetic atoms**

Atoms with unpaired electrons are attracted by a magnet (paramagnetic); atoms with none are not (diamagnetic). Find the number of unpaired electrons of Mn ($Z = 25$), Fe, Zn ($Z = 30$), Cu and Zn^{2+} . Which are diamagnetic? Which ion of manganese, Mn^{2+} or Mn^{3+} , has the most unpaired electrons?

Q15. Challenge ★ **Beyond the known table**

The last element of period 7 is oganesson, Og ($Z = 118$). (a) Write the valence configuration of Og. (b) Using Madelung's rule, which subshells would period 8 fill, and in which order? (Remember that $\ell = 4$ is called g.) (c) Predict the atomic number of the noble gas at the end of period 8.

Worksheet 5 · Atomic properties and periodicity

Data $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$ $c = 3.00 \times 10^8 \text{ m/s}$ $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$ $N_A = 6.022 \times 10^{23} / \text{mol}$
 $1 \text{ D} = 3.336 \times 10^{-30} \text{ C} \cdot \text{m}$. Use Tables 5-1 to 5-4 of the handout (radii, ionisation energies, bond energies, Pauling electronegativities). Calculator allowed.

Q1. Level 1 • Definitions

Define: atomic radius, first ionisation energy, polarisability, electronegativity. For each, say whether it is a property of an isolated atom or of an atom in a bond.

Q2. Level 1 • Average atomic mass of chlorine

Chlorine is 75.77 % ^{35}Cl (34.97 amu) and 24.23 % ^{37}Cl (36.97 amu). Compute its average atomic mass and its molar mass.

Q3. Level 1 • Ranking radii

Using Table 5-1, rank Na, Mg, K and Cl by increasing atomic radius and state the two trends that explain the order.

Q4. Level 1 • Units of ionisation energy

The ionisation energy of sodium is 496 kJ mol^{-1} . Convert it to joules per atom and to electronvolts per atom. Show that 1 eV per atom corresponds to 96.4 kJ mol^{-1} .

Q5. Level 2 •• Mass versus atomic number

- (a) Compute the average atomic mass of potassium: ^{39}K 38.96 amu, 93.26 %; ^{41}K 40.96 amu, 6.73 %.
 (b) Argon (39.95 amu) has $Z = 18$ and potassium $Z = 19$. What problem would appear if elements were ordered by mass? How is it solved today?

Q6. Level 2 •• Explaining the radius trends

Explain, using the valence shell n and the effective nuclear charge, why the radius decreases from Li (157 pm) to F (64 pm) and increases from F to I (133 pm). Why can Ne (154 pm) not be compared with F?

Q7. Level 2 •• An isoelectronic series

Rank O^{2-} , F^- , Na^+ and Mg^{2+} by increasing radius. Justify. Compare also the radius of Na^+ with that of Na, and of Cl^- with that of Cl.

Q8. Level 2 •• The ionisation energies of period 2

The values for period 2 are: Li 520, Be 900, B 801, C 1086, N 1402, O 1314, F 1681, Ne 2081 kJ mol^{-1} .
 (a) State the general trend and explain it. (b) Explain the two exceptions with box diagrams.

Q9. Level 3 ••• Ionising light

- (a) Compute the longest wavelength of a photon able to ionise a sodium atom (496 kJ mol^{-1}), and the one for helium (2372 kJ mol^{-1}). In which region of the spectrum are they? (b) In a PES experiment, UV photons of 21.2 eV hit sodium atoms. What is the kinetic energy of the ejected valence electrons?

Q10. Level 3 ••• Successive ionisation energies

An element of period 3 has successive ionisation energies 738, 1451, 7733, $10\,543 \text{ kJ mol}^{-1}$. (a) Identify it and justify. (b) Why does each ionisation energy exceed the previous one? (c) Which ion does this element form in its compounds?

Q11. Level 3 ●●● Abundance from the average mass

Boron has two isotopes, ^{10}B (10.013 amu) and ^{11}B (11.009 amu), and an average atomic mass of 10.81 amu. Compute the natural abundance of each isotope.

Q12. Level 3 ●●● How polar is H–Cl?

The dipole moment of HCl is 1.08 D and its bond length 127 pm. (a) Compute the partial charge δ (as a fraction of e) carried by each atom. Which atom is δ^- ? (b) Using Table 5-4, rank H–F, H–Cl, H–Br and H–I by increasing polarity.

Q13. Challenge ★ Pauling electronegativity from bond energies

Use Table 5-3 (H–H 436, F–F 158, H–F 567, I–I 151, H–I 299 kJ mol $^{-1}$), the formula $|\chi_A - \chi_B| = 0.102\sqrt{\Delta}$ with $\Delta = D_{AB} - \sqrt{D_{AA}D_{BB}}$, and $\chi_{\text{H}} = 2.20$. (a) Compute χ_{F} . (b) Compute χ_{I} and compare with Table 5-4 (2.66). (c) Why is the H–F bond much stronger than the average of H–H and F–F?

Q14. Challenge ★ The Allred–Rochow scale

(a) Using Slater's rules (Unit 5, "Going further"), show that $Z_{\text{eff}} = 6.1$ for a valence electron of Cl. (b) With $\chi = 0.359 Z_{\text{eff}}/r^2 + 0.744$ (r in Å) and the covalent radius of Cl (99 pm), compute $\chi_{\text{AR}}(\text{Cl})$ and compare with Pauling's 3.16. (c) Without calculation, explain why F has a higher electronegativity than Cl on this scale.

Q15. Challenge ★ Synthesis: an unknown element

Element X is in period 3. Its successive ionisation energies are 578, 1817, 2745 and 11 577 kJ mol $^{-1}$. (a) Identify X and write its configuration with the box diagram of its valence shell. (b) Explain why the first ionisation energy of X is lower than that of Mg (738), although X has one more proton. (c) Compare the radius, the electronegativity and the polarisability of X with those of Mg and Si, using the tables. (d) Which ion does X form? Is it isoelectronic with a noble gas?